W04

GLMs for counts and presence–absence

STATISTICS R PROGRAMMING

Bird counts and presence–absence data do not follow a normal distribution. Generalised linear models (GLMs) pick a suitable distribution (Poisson, negative binomial, binomial) and a link function that keeps predictions within possible values.

DURATION	About 2 hours.
PACKAGES	`tidyverse`, `MASS`, `DHARMa`, `lme4`
FILES	`week04_GLMs.R`, `week04_solutions.R`, `survey_points.csv`, `bird_counts_wide.csv`
DATA	120 visits (40 points x 3) with total birds; woodlark presence per point.

Learning outcomes

- choose a distribution and link for the type of response;
- fit and interpret Poisson and binomial GLMs on the response scale;
- detect overdispersion and use the negative binomial;
- check residuals with `DHARMa` and compare models with AIC;
- account for repeated visits with a mixed model.

1 · Choosing the GLM

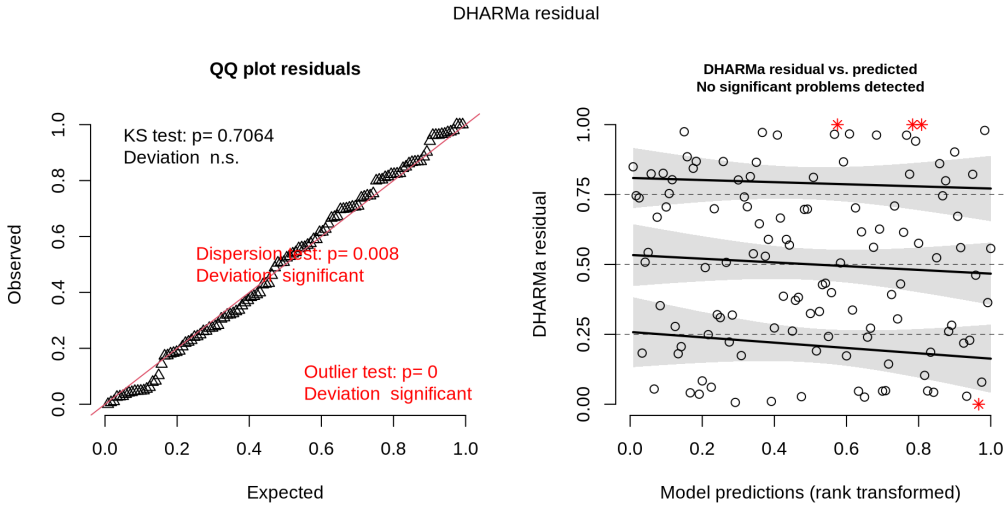
Response	Distribution	Link	Example
Counts (0, 1, 2…)	Poisson	log	birds per visit
Counts with variance > mean	negative binomial	log	birds per visit (overdispersed)
Presence–absence (0/1)	binomial	logit	is woodlark present?
Proportions (successes of n)	binomial	logit	eggs hatched per clutch

With a log link the model is `log(mean) = b0 + b1·x`. So coefficients are interpreted after `exp()`: `exp(b)` is the factor by which the count is multiplied when x increases by one unit.

2 · Poisson and overdispersion

```
m_pois <- glm(total ~ elev_100 + canopy_cover_pct + observer + visit,
              family = poisson, data = visits)
sum(residuals(m_pois, type = "pearson")^2) / df.residual(m_pois)   # 1.52
testDispersion(simulateResiduals(m_pois))                          # p = 0.008
```

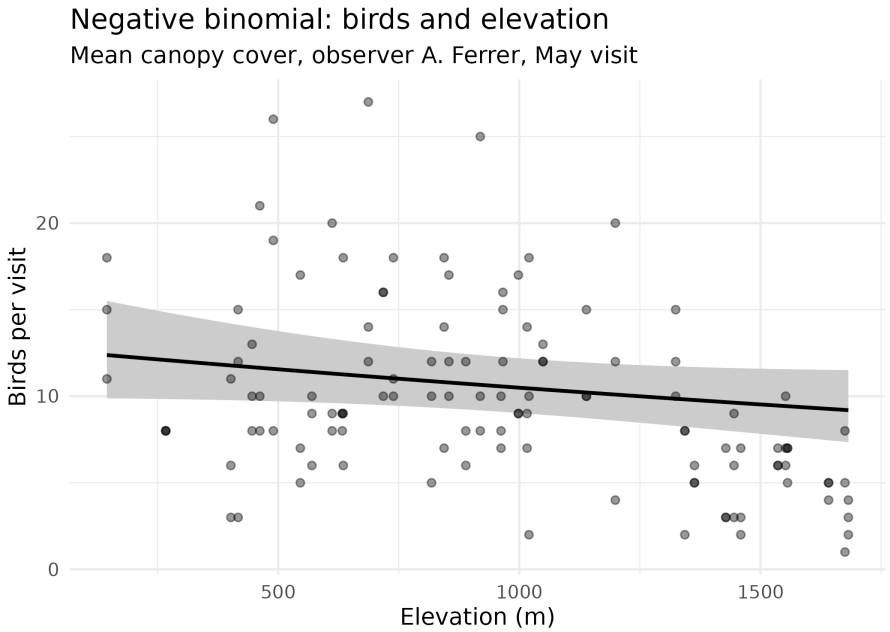
Poisson assumes the variance equals the mean. Here the variance is 50% larger, and the DHARMa test confirms it: standard errors would be too small and we would see "significant" effects that are not. The negative binomial adds a parameter for that extra variation and lowers AIC from 664 to 655.



DHARMa diagnostics for the Poisson model. Left: simulated residuals against the expected distribution. Right: residuals against predictions.

3 · Observer effect

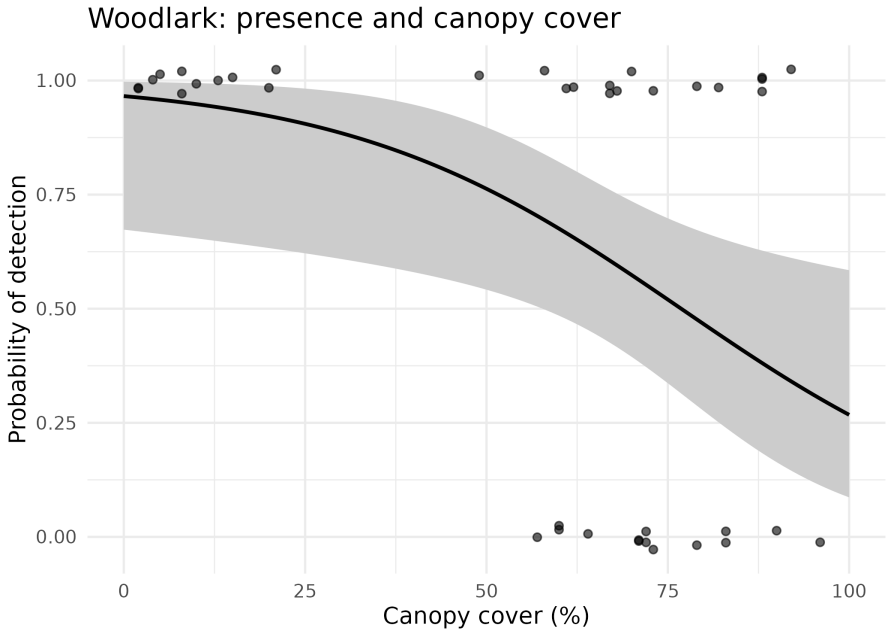
With the negative binomial, dropping observer makes the model clearly worse (likelihood-ratio test, $p < 0.001$). `exp(coef(m_nb))` gives 0.78 for J. Serra: 22% fewer birds than A. Ferrer under the same conditions. Ignoring it would mix differences between people with differences between habitats whenever some observers survey certain sites more.



Negative binomial prediction with its 95% interval, calculated on the log scale and back-transformed so it never goes below zero.

4 · Presence–absence

Woodlark was detected at 26 of the 40 points. The binomial GLM shows detection probability falls with canopy cover: each percentage point multiplies the odds by 0.96, and probability drops below 50% at about 77% cover (exercise 4).



Logistic regression: probability of detecting woodlark against canopy cover. Points are the data (0 or 1).

5 · Repeated visits: mixed models

The three visits to the same point are not independent. `glmer(total ~ ... + (1 | point_id), family = poisson)` gives each point its own average level (random effect). The coefficients are usually similar to the GLM's, but their standard errors are more honest.

Exercises

Answers in `week04_solutions.R`.

1. Poisson GLM for robins per visit with elevation and habitat. Is there overdispersion?
2. By what factor does the number of birds change per 100 m? Give the interval.
3. In the binomial model, replace canopy cover with elevation. Which has the lower AIC?
4. At what canopy cover does probability drop below 50%? (Hint: `-b0 / b1`.)
5. Compare the observer effect in the GLM and the mixed model.

Before Week 5

- Make sure QGIS opens without errors.
- Download the Week 5 data (`montfred.gpkg`, `dem.tif`, `landcover.tif`) into your `data/` folder.